

Work №2. SIMPLE EXAMPLES OF ADAPTIVE ROBUST CONTROLLERS DESIGN

Problem statement. Consider disturbed plant

$$\dot{x} = \theta x + u + \delta, \quad (2.1)$$

where δ is unknown bounded disturbance such that $|\delta(t)| \leq \bar{\delta}$. As in the work №1. x is the state variable, u is the control, θ is the unknown constant parameter.

The objective is to design a control that will provide the boundedness of all the signals in the closed-loop system and ensure the inequality

$$|x_m(t) - x(t)| \leq \Delta, \quad \forall t \geq T, \quad (2.2)$$

where Δ , T are the maximum steady-state error and transient time, respectively, $x_m(t)$ is the output of reference model (1.2). It is assumed that the constants Δ and T can be regulated by the parameters of controller.

Solution № 1. Solution obtained by applying nonlinear static feedback

$$\hat{\theta} = -\gamma x \varepsilon \quad (2.3)$$

in (1.4) instead of adaptation algorithm (1.5). Substituting (2.3) for $\hat{\theta}$ in (1.4) we obtain **robust** control law

$$u = \gamma x^2 \varepsilon - \lambda x + \lambda g. \quad (2.5)$$

Solution № 2. Modify adaptation algorithm (1.5) by applying the leakage factor (static linear feedback):

$$\dot{\hat{\theta}} = -\sigma \hat{\theta} - \gamma x \varepsilon, \quad (2.8)$$

where σ is the constant value. Then use this **adaptive robust** modification together with adjustable control (1.4).

The order of the work

1. Simulate the adaptive stabilizing (reference $g(t) \equiv 0$) system for the linear **disturbed** plant

$$\dot{x} = \theta x + u + \delta$$

consisting of the adjustable controller

$$u = -\lambda x - \hat{\theta} x,$$

adaptation algorithm

$$\begin{aligned} \dot{\hat{\theta}} &= -\gamma x \varepsilon, \\ \varepsilon &= x_m - x = 0 - x = -x, \end{aligned}$$

and the disturbance

$$\delta(t) = (1+t)^{-1/8} \left[1 - \theta(1+t)^{-1/4} - \frac{3}{8}(1+t)^{-5/4} \right].$$

The controller parameters are given as $\gamma = 0.25$, $x(0) = 1$ и $\hat{\theta}(0) = 1$. Separately plot variable x together with x_m , variable u and variable $\tilde{\theta} = \theta - \hat{\theta}$. Make a conclusion.

2. Repeat the previous experiment with the control (2.5) and three different gains γ . Reference signals $g(t)$ are taken from Lab1. Define the influence of the gain γ on the closed-loop system properties. Separately plot variable x together with x_m and variable u .

3. Repeat experiments 2.2 using the control (1.4) with σ -modification (2.8) and with the same three different γ . Repeat experiment for a certain γ and three different coefficients σ . Reference signals $g(t)$ are taken from Lab 1. Define the influence of the coefficients γ and σ on the closed-loop system properties. Separately plot variable x together with x_m , variable u and variable $\tilde{\theta} = \theta - \hat{\theta}$.

Problems and questions +10pts

1. Show that the robust controllers presented in the work ensure the stability of the closed-loop system despite the changes of the parameter θ .

2. Prove the stability of the robust system with the plant (2.1), control (1.4) and nonlinear static feedback

$$\hat{\theta} = -\gamma x \text{sign}(\varepsilon)$$

with a positive adaptation gain γ .

3. Prove the stability of the robust system with the plant (2.1), control (1.4) and the σ -modification of adaptation law

$$\dot{\hat{\theta}} = -\sigma \hat{\theta} - \gamma x \text{sign}(\varepsilon),$$

where γ , σ are positive constants.